Dhiraj Baid

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.86
Scholar English Academy	Class XII (CBSE)	2024	93.20%
Fountainhead School	Class X (IB)	2022	83.90%

ACHIEVEMENTS

Received the **Academic Excellence** award for having the **highest CGPA** in my branch

1st Place - NPCI Hackathon: Won amongst 40+ teams to build a universal reward token exchange platform, prize of *Rs. 1,50,000* sponsored by NPCI

Represented IIT Hyderabad in **InterIIT Cultural Meet 8.0** as Photography Club

EXPERIENCE

Founding Member, Lead Backend Developer

Finstocks AI - AI based Autonomous Trading Platform

- Architected a microservices backend on GCP, Dockerized services with automated container deployment and VM-based SMTP workers, serving 5,000+ users and 10,000+ daily queries at >99.9% uptime.
- Designed the end to end chatbot service architecture from scratch, query parsing, intent-based routing, parallel tool-call dispatch, and dynamic context trimming, reducing token consumption by 80% per query (80% cost reduction).
- Developed HFT signal algorithms in C++ and Go, achieving sub 4 microsecond execution per stock and full 250 stock spike detection sweeps within a 15 minute sliding window in under 5ms, using SIMD aligned floating-point arithmetic and parallel processing.
- Built real-time market data ingestion pipelines and a LangChain agent with parallel tool-call routing for live analysis, delivered results over WebSockets/SSE and automated trade alert and marketing emails via SMTP with zero manual intervention post-deployment.

Freelance Full Stack Developer

School ERP system

- Built end-to-end ERP system for a K-12 school serving 500-1000 users across 8 roles (students, parents, teachers, principal, management, librarian, store staff, bus drivers) as 1 of 3 freelance developers
- Developed 13+ modules spanning payroll, fee collection with payment gateway integration, live GPS bus tracking, library catalogue, attendance, and leave management using a React/Vite frontend and Go (Gin) backend
- Architected MSG91-powered SMS and SMTP as the system's core notification layer, automating absentee alerts, fee due reminders, leave approvals, and bus arrival updates across all 8 user roles.

PROJECTS

x86-64 OS Kernel

- Built a bare-metal x86-64 kernel from scratch; implemented a physical memory manager (PMM) handling a 4GB address space using a bitmap allocator with page-granularity tracking.
- Implemented a full Interrupt Descriptor Table (IDT) for hardware and software interrupt dispatch, a PS/2 keyboard driver, and FAT32 filesystem support.
- Developed an interactive shell with command parsing; all components written without any OS or runtime dependencies.

TCP Load Balancer

- Built a custom Layer 4 TCP load balancer in Go with concurrent connection handling, supporting round-robin, least-connections, and weighted routing across configurable backends with live health monitoring and automatic failover.
- Stress-tested at 10,000 concurrent connections (512-byte payloads) across 3 backends; achieved a 0.19% average failure rate over 5 runs with 5-10s total test duration per run.
- Profiled with pprof; 77.6% of CPU time spent in I/O syscalls, confirming near-zero overhead in application logic.

Mess Registration Portal

Lambda - Dev Club, IIT Hyderabad

- Designed and deployed the backend for the institute-wide mess registration system, handling 5,000+ concurrent users across B.Tech, M.Tech, PhD, and research cohorts.
- Engineered the system to absorb a 5,000-request spike within 5 minutes under a first-come-first-served stampede scenario using Redis queues and PostgreSQL.

POSITIONS OF RESPONSIBILITY

Head Positions: Web Dev Head | International Relations Cell || Head | Infocus - Photography Club

Core Positions: Technical Cell | OCS || Core Developer | Lambda - Dev Club || Core | Infocus - Photography Club

SKILLS

Languages: C, C++, Go, Python, JavaScript, TypeScript, x86-64 Assembly

Frameworks & Libraries: Gin, GORM, FastAPI, LangChain, React, Vite

Infrastructure: PostgreSQL, Redis, Docker, Git, Linux, Google Cloud Platform

Tools: QEMU, GDB, Postman, VS Code, CMake, Makefiles, Beekeeper

Concepts: Low-Latency Systems, Operating Systems, Databases, Concurrent Programming, Memory Management, LLM Optimization, API design, Data Structures and Algorithms, Micro-optimizations

Soft Skills: Communications, Pitching, Event Organizing, Relationship building, Team Collaboration

RELEVANT COURSES

Mathematics: Calculus, Differential Equations, Probability, Elementary Linear Algebra

Computer Science: Introduction to Artificial Intelligence, Introduction to Programming